ETHGLOBAL MUMBAI · HEALTH TRACK

Safe Medicine Ledger

Proving that every batch of medicine was independently tested, before it reaches a patient.

The problem

Medicine makers can skip safety tests or fake test certificates, because nobody outside the factory can check them. That's how a real-looking bottle like Coldrif can contain poison and kill children.

The solution

Independent labs, chosen at random, sign every test result on a tamper-proof blockchain, and a batch can't be sold unless all its tests passed. Pharmacists and patients scan the bottle to see instantly whether it's safe, recalled or fake.

Why this happens today

- The main safety test is done by **the company's own lab**. The company tests its own product and declares it passed.
- Ingredient suppliers send a **certificate on paper or as a PDF**, which is easy to copy or fake.
- Government labs mostly test **random samples after the medicine is already on sale**, often too late.
- Today's QR codes on medicines prove the **box is genuine**, not that **what's inside passed a test**.

What each piece does

PIECE	ITS JOB
Independent lab, picked at random	The company can't test its own product or choose a friendly lab.
Blockchain	A record of test results that nobody can edit or delete.
Smart contract	Automatically blocks any batch without a passing test. A missing test counts as a fail.
QR code on each bottle	Anyone can check the batch's safety, recall status and authenticity in one scan.
Random government re-tests	Catch labs that lie. A lab caught lying loses its approval, and every batch it ever approved is flagged.

Today the company grades its own exam. With this system, a random outside examiner grades it, the marks go into a register nobody can change, the medicine can't be sold without a pass, and anyone can check by scanning the bottle.

EXAMPLE

One bottle of cough syrup

A made-up company, **ABC Pharma**, makes a cough syrup called **KoffEase**. One drum of an ingredient it buys is secretly contaminated with poison (diethylene glycol), as happened with Coldrif.

Without the system (today)

1. The supplier delivers the poisoned ingredient with a **fake certificate** saying "pure, passed testing."
2. ABC Pharma **skips its own test** to save time and money. Nobody outside can tell.
3. Batch #123 is made, bottled and shipped. The bottle and label are **completely genuine**.
4. A mother buys it for her child. The pharmacist has **no way to check** whether it was tested.
5. Children fall sick. The poison is found **weeks later**, and the recall is slow.

With the system

1. Before batch #123 can be sold, the system **randomly assigns an independent lab**. ABC Pharma doesn't get to choose.
2. The lab tests the ingredients and the syrup, **finds the poison**, and records "*Batch #123: FAIL*" on the blockchain with its digital signature. Nobody can change it.
3. The smart contract sees the FAIL and **blocks the batch**. The bottles never get valid QR codes, so no pharmacy accepts them.
4. If ABC Pharma had skipped the test, there would be **no record at all**, and the batch would be blocked just the same.
5. For a safe batch, the pharmacist, or the mother at home, scans the bottle and sees the result:

✅ Safe to sell

KoffEase · Batch #124
Tested by an independent lab: **PASS**
Ingredients tested: **PASS**
Recalled: **No** · Genuine bottle: **Yes**

🛑 Do not sell

KoffEase · Batch #123
Lab test: **FAILED** / no test on record
or: batch recalled by regulator
or: bottle already sold, likely a copy

Result: the poison is stopped at the factory, before it reaches a pharmacy, instead of being discovered after children get sick.

ABC Pharma and KoffEase are made-up names used for illustration. Coldrif (Madhya Pradesh, 2025) is the real-world case that inspired the example: at least 24 children died after taking a cough syrup contaminated with diethylene glycol.